

MA 514: Computer Programming

Course Code	MA 514
Course Name	Computer Programming
L-T-P-C	3-0-0-3
Prerequisites	NA
Intended for	M.Sc./M.S./Ph.D.
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **Introduction to Computer Programming** [4 Lectures]
Programming and Programming Languages, Flowchart, The C Programming Language, Identifiers, Symbolic Constants, Declarations, Arithmetic Operations, Relational and Logical Operations.
- **Branching and Iteration** [6 Lectures]
If-Else, ?: Conditional Expression, Switch, While Loops, Do-While Loops, For Loops, Break and Continue, Goto.
- **Functions** [4 Lectures]
Function Prototypes, Call by reference, Call by arguments, recursive function, inline function.
- **Pointers** [6 Lectures]
What is a Pointer? Pointer Syntax, Pointers and Arrays, Pointer Arithmetic, Return Values and Pointer, Pointers to Pointers, Function Pointers, Dynamic Memory allocation.
- **Arrays and Strings** [5 Lectures]
Formatted IO: printf, scanf, string formatting; File IO: Opening and Closing Files, Standard IO, Sequential File Operations.
- **Object-oriented programming** [12 Lectures]
Introduction to User define datatype, Fundamentals of the object-oriented approach, introduction to class and its components, constructors, referring to objects of a class, static members, classes and their friends, Introduction to STL and application.

Text Books

1. V. Rajaraman, **COMPUTER PROGRAMMING IN C**, PHI Learning, 2004.
2. E. Balagurusamy, **Programming In Ansi C**, 3rd Edition, Tata McGraw-Hill Publication, New Delhi, 2004.
3. Walter Savitch, **Problem Solving with C++**: Global Edition, 9th Edition, Pearson Education, 2014.
4. Robert Lafore, **Object Oriented Programming In C++**, 4th Edition, Pearson Education India, 2004.

References

1. Bjarne Stroustrup, **The C++ Programming Language**, 4th Edition, Pearson Education, 2013.
2. Brian W. Kernighan, **The C Programming Language (Ansi C Version)**, 2nd Edition, PHI, 1990.
3. Brian W. Kernighan, Dennis M. Ritchie, **Programming Languages C with Practicals**, 1st Edition, Margham Publications, 2012.